INVENTION BRIEF / RECREATED EDITION

PREPARED FOR MICHAEL GEE

Room conditions, made visible.



BIRP is a proposed wall-mounted environmental monitor with a round display and a physical bird that travels around its perimeter. Better conditions place the bird toward the top; worsening conditions move it toward the bottom. A light halo and readable measurements explain the signal.

CURRENT STAGE: DOCUMENTED CONCEPT + BROWSER DEMONSTRATION

01 / THE SIGNATURE INTERFACE

A physical perimeter bird makes the room-condition signal visible at a glance. The founder specified a 5-inch round display; final enclosure dimensions and the motion mechanism remain open.

02 / STANDALONE BY DESIGN

Core room sensing and local feedback are intended to work without an app or subscription, after required sensor startup. The display provides the reason for a state change.

03 / OPTIONAL CONNECTED SERVICES

The broader vision adds app history, remote monitoring, smart-home integration, video intercom, and multiple room units with a master-unit role. These features are not implemented.

COLLABORATION GOAL

Make the interaction understandable, review the design with suitable engineering and industrial-design partners, and define an achievable first physical proof. The initial pitch focuses on home-room air awareness; customer demand is still to be tested.

How the room becomes a signal.

The bird's location around the display is the primary indicator. Body tilt is a separate possible gesture; it must not replace or obscure perimeter position.



TOP
Better conditions

SIDE
Attention preset

BOTTOM
Worse conditions

Illustrative positions only. The founder specified top/bottom meaning; the side position is a prototype preset. No measured thresholds are assigned to these drawings.

WHEN A READING IS MISSING

Show an explicit unknown status and a violet halo. Retain the last referenced bird position where safe, clearly identified as historical. Missing data must never produce a reassuring all-clear signal.

PROPOSED FIRST-BENCH SIGNAL FLOW

MEASURE
CO2 / PM2.5
Temperature / humidity

→

INTERPRET
Validate freshness
Classify channels separately

→

EXPRESS
Perimeter bird
Halo + cause on display

Sensor-specific readiness, classification policy, position referencing, physical travel limits, and fault behavior must be selected and tested before hardware performance can be claimed.

The device, in context.

The original exploded illustration records the visual direction and optional modules. It is concept art: component labels, dimensions, fit, and internal layer details are not validated engineering specifications.



Original project artwork supplied by the founder. The perimeter-motion requirement in the written product vision remains authoritative where the illustration is ambiguous.

AREA	PROPOSED FIRST EXPERIMENT	BROADER INTENDED PRODUCT
Display and bird	Round display, perimeter bird, halo, and cause-aware feedback.	5-inch display target; final enclosure and mechanism remain unselected.
Sensing	CO2, PM2.5, temperature, and humidity.	Final sensor suite and product policy remain open.
Connectivity	Core sensing and local output independent of app setup.	Optional app/services, smart-home integration, video intercom, and multi-room/master-unit roles.

The path to first physical proof.

1 **Test the need and the interaction.**

Complete the concept packet, then ask prospective users to explain the bird signal and the problem they would use it to solve. Record actual feedback for one initial audience.

2 **Review a buildable architecture.**

With suitable resources or a partner, review display and sensor candidates, full orbital travel, gravity loading, position reference, power, heat, and noise. The same-hardware goal needs a separate feasibility review.

3 **Demonstrate the complete loop.**

Show sensor readings, independent channel classification, cause text, halo, perimeter movement, missing-reading status, and qualified recovery. Record the build configuration and measured results.

4 **Choose the product-development path.**

Use the evidence to assess partner-led development, a licensing discussion, or another resource-appropriate prototype step. Camera/audio, app services, and multi-room roles require further architecture work.

WHAT A PARTNER CAN HELP EVALUATE

Mechanical feasibility, sensor integration, industrial design, user understanding, and a costed prototype plan. No purchase, funding commitment, production price, or partnership is represented by this brief.

SOURCE MATERIAL AND CURRENT PROJECT

Original design conversation - supplied by the founder

Minimal founder plan - supplied by the founder

Current product vision

Control draft 0.2

BIRP website

GitHub repository

This is a newly recreated brief, not the missing original PDF. Current evidence consists of project documents, two original visual references, and a browser demonstration. A physical device, customer demand, manufacturing cost, and operating performance have not been validated. Smoke/CO integration remains outside the proposed bench scope.